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Portable air purifier

Abstract

A portable air purifier is provided. In the portable air purifier, a fan base is disposed between a fan module including a mixed flow fan and a filter, and forms a passage that connects the filter and the fan module while coupling the filter and the fan module. Accordingly, the portable air purifier can ensure improvement in air purification performance with no increase in size of the portable air purifier.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6017375	12/1999	Duell	55/497	F24F 1/0071
7998231	12/2010	Zheng	55/467	B01D 46/62
2005/0193803	12/2004	Carlson et al.	N/A	N/A
2007/0221061	12/2006	Steiner	55/467	F24F 8/10
2015/0352480	12/2014	Morison	96/417	B01D 46/0086
2023/0211269	12/2022	Hsieh	96/226	B01D 46/0005

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1752618	12/2005	CN	N/A
203454279	12/2013	CN	N/A
205332377	12/2015	CN	N/A
205468438	12/2015	CN	N/A
206540257	12/2016	CN	N/A
3385625	12/2021	EP	B01D 46/42
10-2017-0138289	12/2016	KR	N/A
10-2027302	12/2018	KR	N/A
10-2032401	12/2018	KR	N/A
10-2055522	12/2019	KR	N/A

OTHER PUBLICATIONS

German Office Action dated Nov. 30, 2022. cited by applicant

International Search Report dated Oct. 13, 2020 issued in Application No. PCT/KR2020/005390. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS

(1) This application is a U.S. National Stage Application under 35 U.S.C. § 371 of PCT Application No. PCT/KR2020/005390, filed Apr. 23, 2020, which claims priority to Korean Patent Application No. 10-2020-0013181, filed Feb. 4, 2020, whose entire disclosures are hereby incorporated by reference.

TECHNICAL FIELD

(2) Disclosed herein is a portable air purifier, and specifically, a portable air purifier that can be carried for use.

BACKGROUND ART

(3) Air purifiers are widely used in our daily lives. The devices can filter physical particles such as dust, fine dust, ultra fine dust and the like, chemical substances such as odorant particles, harmful gases and the like, and microorganisms such as germs, viruses and the like, in air, to purify the air.

(4) People cannot live without air purifiers in an industrial society since more and more people are greatly affected by fine dust and suffer from allergies. Accordingly, there is a growing demand for the device.

(5) Ordinarily, a large-sized air purifier is used in a house that is 100 square meters or greater. The air purifier can be provided with a filter for physical particles such as dust and the like, a filter for chemical substances such as gas and the like, and a filter for microorganisms such as germs, viruses and the like, in combination. That is, such a large-sized air purifier capable of accommodating various types of filters can be used in a large space.

(6) However, air purifiers are rarely used in a narrow space such as a studio apartment, a space in a vehicle and the like, or in a very wide space such as a public library and the like or an outdoor space, considering space availability, portability and energy efficiency. Additionally, a user who moves from place to place usually uses an air purifier small enough to carry. Under the circumstances, there is a growing need for a portable air purifier that is easy to carry for use.

(7) Portable air purifiers need to be small and lightweight enough for users to carry such that the users can easily carry and use the portable air purifiers anywhere. That is, the devices are useful for people who often go out and move from place to place instead of staying in a place such as a house.

(8) The amount of purified air discharged from a portable air purifier is less than the amount of purified air discharged from an ordinary air purifier that is installed in a place. Additionally, the portable air purifier discharges purified forward in a narrower range than the ordinary air purifier. The portable air purifier has the above problems inherently, since the portable air purifier has a small size. As the size of the portable air purifier decreases, it is difficult to increase the amount of purified air discharged and the range of forward discharge of purified air.

(9) A decrease in the amount of discharge of purified air and the range of forward discharge of purified air of the portable air purifier makes it difficult for air purified by the portable air purifier to reach a user, in particular, a user's face. The portable air purifier's failure in reaching the user's face means that the portable air purifier cannot provide air purification performance properly.

DESCRIPTION OF INVENTION

Technical Problems

(10) One objective of the present disclosure is to provide a portable air purifier that is compact and lightweight, and has an improved structure enabling purified air to reach a user's face effectively.

(11) Another objective of the present disclosure is to provide a portable air purifier that has an improved structure capable of suppressing the occurrence of flow loss caused by whirlwind.

(12) Yet another objective of the present disclosure is to provide a portable air purifier that has an improved structure capable of reducing the possibility of wear or damage to a fan and its surrounding components.

Technical Solutions

(13) In a portable air purifier of one embodiment, a fan base is disposed between a fan module including a mixed flow fan and a filter, and forms a passage that connects between the filter and the fan module while coupling the filter and the fan module.

(14) In a portable air purifier of another embodiment, a fan cover and a fan base are disposed at one side and the other side of a fan module including a mixed flow fan-type fan in the axial direction, the fan base includes a bell mouth protruding toward the fan module, and at least a portion of a fan is inserted into an air discharge part being formed at the fan cover in a way that penetrates the fan cover.

- (15) Accordingly, in the portable air purifier, the size of the fan may increase, or the size of the portable air purifier may decrease as much as space occupied by a guide vane decreases.
- (16) Additionally, in the portable air purifier having the above structure, since at least a portion of the fan is inserted into the fan cover, the entire size of the portable air purifier may not increase although the bell mouth is added to guide a suction flow.
- (17) In a portable air purifier of another embodiment, a projection rib is disposed outside a bell mouth in the radial direction of the bell mouth, and is spaced a predetermined distance apart from a shroud while protruding toward the shroud in the lateral direction and forms a slant surface parallel with a slant surface of the shroud.
- (18) Accordingly, flow loss caused by whirlwind may be suppressed, and the rigidity of the fan base may increase, suppressing the deformation of the fan base, caused by an external force.
- (19) A portable air purifier according to one aspect may include: a case having a suction surface on one side thereof in the axial direction and having a discharge surface on the other side thereof in the axial direction; a filter being disposed between the suction surface and the discharge surface; a fan module being disposed between the discharge surface and the filter, and suctioning air, having passed through the filter, and discharging the air to the discharge surface; and a fan base being disposed between the filter and the fan module, being coupled to the fan module, and forming a passage that connects between the filter and the fan module.
- (20) The fan base includes a base plate being disposed between the filter and the fan module; and a fan inlet being formed at the base plate in a way that penetrates the base plate, and forming a passage that connects between the filter and the fan module, for example.
- (21) The fan base further includes a bell mouth protruding from the base plate in the axial direction, and the fan inlet is formed at the inside of the bell mouth in a way that penetrates in the radial direction of the bell mouth, for example.
- (22) At least a portion of the bell mouth is disposed to overlap the fan module, for example.
- (23) The fan module includes a fan being rotatably disposed between the bell mouth and the discharge surface, the bell mouth has a first surface that faces the filter module and is convex toward the filter module and has a second surface that faces the fan module and is depressed, while protruding from the inner circumferential surface of the base plate in the axial direction, and at least a portion of the fan is inserted into space surrounded by the second surface, for example.
- (24) The fan includes a hub being rotatably provided, a shroud being spaced a predetermined distance apart from the hub and being disposed outside the hub in the radial direction of the hub, and a fan blade protruding from the hub in the centrifugal direction and connecting between the hub and the shroud, and at least a portion of the shroud is inserted into the space surrounded by the second surface, for example.
- (25) The shroud forms a slant surface that inclines in the centrifugal direction as the shroud becomes farther from the filter module in the axial direction, and the fan base further includes a projection rib that protrudes from the base plate toward the shroud in the axial direction while surrounding the bell mouth from the outside in the radial direction, for example.
- (26) The projection rib is spaced a predetermined distance apart from the shroud, and forms a slant surface parallel with the slant surface of the shroud, for example.
- (27) The projection rib forms a slant surface that extends in the centrifugal direction from an end portion outside the bell mouth in the radial direction of the bell mouth, for example.
- (28) The base plate, the bell mouth and the projection rib are integrally formed, for example.
- (29) The direction in which the fan suctions air is the same as the direction in which the suction surface and the discharge surface are disposed, for example.
- (30) The portable air purifier further includes a fan cover being disposed between the discharge surface and the fan module, the fan cover is provided with an air discharge part forming a passage that connects between the discharge surface and the fan, and the fan module is disposed to face the air discharge part, for example.

- (31) The air discharge part is formed at the fan cover in a way that penetrates in the axial direction, and at least a portion of the fan module is inserted into the air discharge part, for example.
- (32) The fan base further includes a bell mouth that protrudes from the base plate in the axial direction, and at least a portion of the bell mouth is disposed to overlap the fan module, for example.
- (33) The fan module includes: a motor including a stator and a rotor that is disposed outside the stator in the radial direction of the stator; a fan case supporting at least a portion of the motor; and a fan rotating together with the rotor, the fan case is coupled to the fan base, and the fan base is coupled to the filter, for example.
- (34) The fan base includes a base plate being disposed between the filter and the fan module, and a fastening projection protruding from the base plate toward the fan case, and the fan module includes a projection boss protruding from the fan case toward the fan base and being coupled to the fastening projection, for example.
- (35) The portable air purifier further includes a fan cover being disposed between the discharge surface and the fan module, and the fan cover includes a fastening projection protruding toward the fan module and being coupled to the fan case, for example.
- (36) Additionally, a fan outlet is formed at the fan case in a way that penetrates the fan case, the fan cover is provided with an air discharge part forming a passage that connects between the discharge surface and the fan outlet, and as the fastening projection is coupled to the fan case, the fan outlet and the air outlet are disposed to overlap each other in the axial direction, for example.
- (37) The fan module includes a mixed flow fan that suctions air having passed through the filter in the axial direction and discharges the air in a direction between the axial direction and the radial direction, for example.

Advantageous Effects

- (38) The portable air purifier according to the present disclosure is provided with a fan that is advantageous in discharging high-pressure high-speed air far away without a structure such as a guide vane, such that the size of the fan increases or the size of the portable air purifier decreases as much as space occupied by the guide vane decreases.
- (39) Thus, according to the present disclosure, air purification performance improves further, with no increase in the size of the portable air purifier, caused by the fan.
- (40) According to the present disclosure, flow loss caused by whirlwind is suppressed, noise is reduced, and a high level of suction and discharge performance is provided, ensuring improvement in air purification performance.
- (41) According to the present disclosure, the occurrence of contact between a fan and a fan base, caused by an external force, is effectively suppressed, reducing the possibility of wear or damage to the fan and its surrounding components and ensuring improvement in durability of a portable air purifier.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a front perspective view showing the front surface side of a portable air purifier of one embodiment.
- (2) FIG. 2 is an exploded perspective view showing a state in which the portable air purifier in FIG. 1 is exploded.
- (3) FIG. 3 is rear perspective view showing the rear surface side of the portable air purifier in FIG. 1.
- (4) FIG. 4 is a cross-sectional view along line “IV-IV” in FIG. 1.
- (5) FIG. 5 is a front perspective view separately showing a case in FIG. 1.

- (6) FIG. 6 is a rear perspective view separately showing the case in FIG. 1.
- (7) FIG. 7 is a cross-sectional view specifically showing the structure of portion “VII” in FIG. 4.
- (8) FIG. 8 is a perspective view separately showing a fan module of one embodiment.
- (9) FIG. 9 is a perspective view separately showing a fan case in FIG. 8
- (10) FIG. 10 is a front view separately showing the fan in FIG. 8.
- (11) FIG. 11 is a side view showing the fan in FIG. 10.
- (12) FIG. 12 is a cross-sectional view along line “X II-X II” in FIG. 11.
- (13) FIG. 13 is an exploded perspective view separately showing a fan cover and a fan module.
- (14) FIG. 14 is an exploded perspective view separately showing a fan module, a fan base and a filter.
- (15) FIG. 15 is a rear perspective view showing the rear surface of a fan base.
- (16) FIG. 16 is a perspective view showing a coupling state between the fan base and the filter.
- (17) FIG. 17 is a view showing an aspect of the air flow of the portable air purifier of one embodiment.
- (18) FIG. 18 is a view showing the structure of a fan base provided in a portable air purifier of another embodiment.
- (19) FIG. 19 is a view showing the structure of a fan base without a projection rib.
- (20) FIG. 20 is a cross-sectional view showing the structures of the fan and the fan base in FIG. 19.
- (21) FIG. 21 is a cross-sectional view showing the structures of the fan and the fan base in FIG. 18.
- (22) FIG. 22 is a table showing results of measurement of noise and press volume, with or without a projection rib.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS

(23) The above-described aspects, features and advantages are specifically described hereunder with reference to the accompanying drawings such that one having ordinary skill in the art to which the present disclosure pertains can easily embody the technical spirit of the disclosure. In the disclosure, detailed description of known technologies in relation to the subject matter of the disclosure is omitted if it is deemed to make the gist of the disclosure unnecessarily vague. Below, preferred embodiments according to the disclosure are specifically described with reference to the accompanying drawings. In the drawings, identical reference numerals can denote identical or similar components.

- (24) The terms “first”, “second” and the like are used herein only to distinguish one component from another component. Thus, the components should not be limited by the terms. Certainly, a first component can be a second component unless stated to the contrary.
- (25) When one component is described as being “in the upper portion (or the lower portion)” or “on (or under)” another component, one component can be directly on (or under) another component, and an additional component can be interposed between the two components.
- (26) When any one component is described as being “connected”, “coupled”, or “connected” to another component, any one component can be directly connected or coupled to another component, but an additional component can be “interposed” between the two components or the two components can be “connected”, “coupled”, or “connected” by an additional component.
- (27) Throughout the disclosure, each component can be provided as a single one or a plurality of ones, unless stated to the contrary.
- (28) In the disclosure, singular forms include plural forms as well, unless explicitly indicated otherwise. It is to be understood that the terms such as “comprise” or “include” and the like, when used in this disclosure, are not interpreted as necessarily including stated components or steps, but can be interpreted as excluding some of the stated components or steps or as further including additional components or steps.
- (29) Throughout the disclosure, the terms “A and/or B” as used herein can denote A, B or A and B, and the terms “C to D” can denote C or greater and D or less, unless stated to the contrary.
- (30) [Exterior of Portable Air Purifier]

(31) FIG. 1 is a front perspective view showing the front surface side of a portable air purifier of one embodiment, and FIG. 2 is an exploded perspective view showing a state in which the portable air purifier in FIG. 1 is exploded. FIG. 3 is rear perspective view showing the rear surface side of the portable air purifier in FIG. 1, and FIG. 4 is a cross-sectional view along line “IV-IV” in FIG. 1.

(32) In the embodiment, a portable air purifier 50 is described as an example.

(33) Referring to FIGS. 1 to 4, the portable air purifier 50 in the embodiment may be approximately formed into a cuboid. The portable air purifier 50 includes a case 520, a front surface panel 510 and a rear surface panel 570.

(34) The case 520 forms the skeleton of the exterior of the portable air purifier 50. Various types of components are accommodated in the case 520.

(35) Both sides of the case 520 in a first direction may be open. That is, the front and the rear of the case 520 may be open. A suction surface may be disposed on one side of the case 520 in the first direction, and a discharge surface may be disposed on the other side of the case 520 in the first direction.

(36) For example, the suction surface may be disposed at the rear of the case 520, and the rear surface panel 570 may be disposed at the rear of the case 520, where the suction surface is disposed. Additionally, the discharge surface may be disposed at the front of the case 520, and the front surface panel 510 may be disposed at the front of the case 520, where the discharge surface is disposed.

(37) The suction surface may indicate a virtual surface corresponding to the boundary between the inside of the case 520 and the outside of the case 520, at the rear of the case 520. The discharge surface indicates a virtual surface corresponding to the boundary between the inside of the case 520 and the outside of the case 520, at the front of the case 520.

(38) In the embodiment, the suction surface and the discharge surface may actually be openings formed at the rear and the front of the case 520, since both the rear and the front of the case 520 are open. In the embodiment, the suction surface and the discharge surface are disposed in parallel, for example.

(39) The front surface panel 510 is coupled to the front of the case 520. The front surface panel 510 forms the exterior of the front surface of the portable air purifier 50.

(40) The rear surface panel 570 is coupled to the rear of the case 520. The rear surface panel 570 forms the exterior of the rear surface of the portable air purifier 50.

(41) The portable air purifier 50 may be entirely formed into a standing cuboid that is elongated in the up-down direction. Accordingly, a user may use the portable air purifier 50 in the state in which the portable air purifier 50 stands or lies. Additionally, the portable air purifier 50 may stay in the same position reliably without rolling in a moving vehicle, even if the portable air purifier 50 is used in the state in which the portable air purifier 50 lies.

(42) Directions are defined as follows. A direction from the case 520 to the front surface panel 510 is referred to as a front, and a direction from the case 520 to the rear surface panel 570 is referred to as a rear. At this time, a “first direction” denotes the front-rear direction. Additionally, the first direction may have the same meaning as the axial direction. The axial direction may be defined as a parallel direction with the lengthwise direction of a shaft disposed at a fan module 540 that is described below. A “second direction” is a direction perpendicular to the first direction and denotes the left-right direction. A “third direction” is a direction perpendicular to the first direction and the second direction and denotes the up-down direction.

(43) [Entire Structure of Portable Air Purifier]

(44) The portable air purifier 50 in the embodiment includes a front surface panel 510, a case 520, a fan cover 530, a fan module 540, 545, a filter module 550, a battery 560, a rear surface panel 570 and a rear surface cover 580.

(45) The front surface panel 510 is disposed at the front most side of the portable air purifier 50 and forms the exterior of the front surface of the portable air purifier 50. Air purified by the portable air

purifier **50** is discharged outward through the front surface panel **510**. To this end, a plurality of outlets **510a** is provided on the front surface panel **510**.

(46) The case **520** forms the skeleton of the exterior of the portable air purifier **50**. The exteriors of the upper surface, lateral surfaces and lower surface of the portable air purifier **50** are formed by the case **520**. Accommodation space is formed in the case **520**. Various types of components such as a fan cover **530**, a fan module **540**, **545**, a battery **550**, a filter module **560** and the like, constituting the portable air purifier **50**, are accommodated in the accommodation space. The case **520** has enough strength to protect the accommodated components from an external impact, for example.

(47) The fan cover **530** is accommodated in the accommodation space of the case **520** and disposed at the front of the fan module **540**, **545**. That is, the fan cover **530** is disposed between the front surface panel **510** and the fan module **540**, **545**, inside the case **520**.

(48) The fan cover **530** fixes the fan module **540**, **545** to the inside of the case **520**.

(49) Additionally, the fan cover **530** also induces air being blown by the fan module **540**, **545** to flow straight toward the front, without causing the spread of the air to surrounding areas. Further, the fan cover **530** may be involved in fixing the filter module **550** and the battery **560**.

(50) The fan module **540**, **545** may be accommodated in the accommodation space of the case **520**, and disposed between the discharge surface and the filter module **550**. Specifically, the fan module **540**, **545** may be disposed between the fan cover **530** and the filter module **550**. That is, the fan module **540**, **545** is disposed at the rear of the fan cover **530** and the front of the filter module **550**. The fan module **540**, **545** suctions air from the rear of the portable air purifier **50** and discharges air to the front of the portable air purifier **50**.

(51) In the embodiment, the fan module **540**, **545** includes a mixed flow fan respectively, for example. The fan module **540**, **545** may suction air having passed through the filter module **550** in the axial direction and discharge the air in a direction between the axial direction and the radial direction.

(52) The filter module **550** is accommodated in the accommodating space of the case **520** and disposed between the fan module **540**, **545** and the rear surface panel **570**. That is, the filter module **550** is disposed at the rear of the fan module **540**, **545** and the front of the rear surface panel **570**.

(53) The filter module **550** purifies air suctioned through the rear of the portable air purifier **50**. The air that is purified while passing through the filter module **550** passes through the fan module **540**, **545**, the fan cover **530** and the front surface panel **510** and then is discharged from the front of the portable air purifier **50**.

(54) The filter module **550** may include a filter case **551** and a filter **559**.

(55) The filter case **551** forms the skeleton of the exterior of the filter module **550**. In the embodiment, the filter case **551** is formed into a cuboid, the rear surface of which is open, for example. Insertion space for accommodating the filter **559** is formed in the filter case **551**. The rear of the filter case **551** is open. Accordingly, a passage for inserting the filter **559** into insertion space in a case main body part **552** is formed.

(56) The filter **559** is mounted in the insertion space of the filter case **551**. The filter case **551** may be provided with a mounting groove or a mounting projection that allows the filter **559** to be firmly mounted in the filter case **551**, on the inner surface thereof.

(57) Additionally, the filter case **551** is provided with a penetration hole **552** that forms a passage between the insertion space and the fan module **540**, **545**. The penetration hole **552** is formed in a way that penetrates in the front-rear direction on the front surface of the filter case **551**. The penetration hole **552** forms a passage for allowing air having passed through the filter **559** to flow toward the fan module **540**, **545**.

(58) A plurality of penetration holes **552** is formed on the front surface of the filter case **551**, and each of the penetration holes **552** is formed in a way that penetrates in the front-rear direction on the front surface of the filter case **551**.

(59) In the embodiment, a plurality of penetration holes **552** is provided on the front surface of the filter case **551**, and each penetration hole **552** is formed into a hexagon, for example. The plurality of penetration holes **552** is arranged in the form of a honeycomb. Accordingly, a honeycomb structure may be formed on the front surface of the filter case **551**.

(60) The honeycomb structure, formed on the front surface of the filter case **551** as described above, ensures improvement in the rigidity of the filter case **551**, makes the filter case **551** lightweight as well as ensuring a passage for allowing air to flow.

(61) The battery **560** is accommodated in the accommodation space of the case **520**, and disposed under the fan module **540**, **545** and the filter module **550**. The battery **560** can supply a power source for driving the portable air purifier **50**. To this end, the battery **560** may electrically connect to the fan module **540**, **545**, the filter module **550** and at least one of a sub printed circuit board (PCB) **590** and a main PCB **595** that are described hereafter.

(62) Together with the rear surface cover **580**, the rear surface panel **570** is disposed at the rearmost side of the case **520** and forms the exterior of the rear surface of the portable air purifier **50**. The rear surface panel **570** is disposed behind the filter module **560**. External air is suctioned into the portable air purifier **50** through the rear surface panel **570**. To this end, the rear surface panel **570** has a plurality of first inlets **570a**, thereon.

(63) Additionally, together with the rear surface panel **570**, the rear surface cover **580** is disposed at the rearmost side of the case **520** and forms the exterior of the rear surface of the portable air purifier **50**. The rear surface cover **580** is disposed behind the battery **560**. In the embodiment, the area behind the filter module **550** is covered by the rear surface panel **570**.

(64) The area behind the battery **560** is covered by the rear surface cover **580**.

(65) The rear surface cover **580** may have a second inlet **580a**. The second inlet **580a** is formed in a way that penetrates on the rear surface cover **580**. The second inlet **580a** forms a passage connecting among the rear of the case **520**, the battery **560** and a sensor module **600**. Through the second inlet **580a**, external air may flow to the battery **560** and the sensor module **600** in a second area B.

(66) [Structure of Case]

(67) FIG. **5** is a front perspective view separately showing a case in FIG. **1**, and FIG. **6** is a rear perspective view separately showing the case in FIG. **1**.

(68) Referring to FIGS. **3**, **5** and **6**, the portable air purifier **50** includes a case **520** that forms the skeleton of the exterior of the portable air purifier **50**. The case **520** has accommodation space therein, and one side and the other side of the accommodation space in the first direction are open.

(69) Directions are defined as follows. A direction from the case **520** to the front surface panel **510** is referred to as a front, and a direction from the case **520** to the rear surface panel **570** is referred to as a rear. At this time, a “first direction” denotes the front-rear direction. A “second direction” is a direction perpendicular to the first direction and denotes the left-right direction. A “third direction” is a direction perpendicular to the first direction and the second direction and denotes the up-down direction.

(70) In the embodiment, the case **520** is formed into a cuboid having the front surface and the rear surface that are open, and therein, has accommodation space having the front and the rear that are open. The case **520** may be made of a metallic material. In the embodiment, the case **520** is made of a material such as aluminum that is lightweight and has high strength, for example.

(71) The rear surface of the case **520** is open to suction external air. The front surface of the case **520** is open to discharge air purified in the accommodation space of the case **520**. Through the open front surface and the open rear surface of the case **520**, various types of components constituting the portable air purifier **50** may be installed in the accommodation space of the case **520**.

(72) A filter module **550** may be mounted in the case **520** or detached from the case **520** through the open rear surface of the case **520**. The rear surface panel **570** is coupled to the open rear surface of the case **520**. The rear surface panel **570** coupled to the case **520** covers the open rear surface of

the case **520**.

(73) The case **520** may include a first surface part **521**, a second surface part **523**, a first connection surface part **525**, a third surface part **527**, and a second connection surface part **529**, while being formed into a cuboid having the front surface and the rear surface that are open.

(74) The first surface part **521** forms a surface in a direction that is perpendicular to the second direction perpendicular to the first direction. That is, the first surface part **521** forms a lateral surface of the case **520**. The first surface part **521** is formed into a vertical flat surface that forms a wall covering a lateral portion of the accommodation space in the case **520**.

(75) The case **520** has a pair of first surface parts **521**, and the first surface parts **521** face each other and are spaced a predetermined distance apart from each other. At this time, the pair of first surface parts **521** is disposed in the second direction, i.e., a direction in which the first surface parts **521** are disposed side by side in the left-right direction.

(76) The second surface part **523** forms a surface in a direction that is perpendicular to the third direction and is disposed at one side of the pair of first surface parts **521** in the third direction. That is, the second surface part **523** is disposed in the upper portion of the first surface part **521**, and forms a flat surface in a direction parallel with the direction in which the first surface parts **521** are spaced, i.e., a horizontal flat surface. The second surface part **523** forms the upper surface of the case **520**.

(77) The first connection surface part **525** is disposed between the first surface part **521** and the second surface part **523**. The first connection surface part **525** is disposed respectively between the end portion of one side of the second surface part **523** and the first surface part **521** below the end portion of one side of the second surface part **523**, and between the end portion of the other side of the second surface part **523** and the first surface part **521** below the end portion of the other side of the second surface part **523**.

(78) Each of the first connection surface parts **525** connects between the first surface part **521** and the second surface part **523** in a rounded manner. The first connection surface part **525** makes an upper edge of the case **520**, at which the first surface part **521** and the second surface part **523** connect, round, improving the safety of a product and the aesthetic qualities of the exterior of a product.

(79) The third surface part **527** is disposed below the second surface part **523** and the first surface part **521**, and forms a flat surface parallel with the second surface part **523**. The third surface part **527** forms the lower surface of the case **520**. Additionally, the third surface part **527** is a portion that supports the portable air purifier **50** such that the portable air purifier **50** keeps standing.

(80) The second connection surface part **529** is disposed between the first surface part **521** and the third surface part **527**. The second connection surface part **529** is disposed respectively between the end portion of one side of the third surface part **527** and the first surface part **521** over the end portion of one side of the third surface part **527**, and between the end portion of the other side of the third surface part **527** and the first surface part **521** over the end portion of the other side of the third surface part **527**.

(81) Each of the first connection surface parts **525** connects between the first surface part **521** and the second surface part **523** in a rounded manner. The first connection surface part **525** makes an upper edge of the case **520**, at which the first surface part **521** and the third surface part **527** connect, round, improving the safety of a product and the aesthetic qualities of the exterior of a product.

(82) The case **520** may have a power button **592** in the upper portion thereof, i.e., in the second surface part **523** of the case **520**. The power button **592** is provided as a manipulation button for turning on/off the portable air purifier **50**.

(83) Additionally, the case **520** may have an air volume control button **593**, in the upper portion thereof. The air volume control button **593** is provided as a manipulation button for increasing or decreasing the air volume of the portable air purifier **50**. The air volume control button **593** may be

disposed near the power button **592** such that the user identifies and manipulates the air volume control button **593** readily.

(84) Further, the case **520** may have a strap mounting part **505** for coupling the strap **501** to the case **520**. The strap **501** is provided to allow the user to hold the case **520**. The user can move up the portable air purifier **50** by holding the strap **501** without directly holding the portable air purifier **50**. That is, the strap **501** helps to improve the portability of the portable air purifier **50**.

(85) The accommodation space in the case **520** may be divided into a first area A and a second area B. When the accommodation space is divided in the up-down direction, the upper area is the first area A, and an area under the first area A is the second area B. The first area A and the second area B are divided conceptually rather than physically.

(86) In the embodiment, among the components accommodated in the accommodation space, the fan modules **540**, **545** and the filter module **550** are disposed in the first area A, and the battery **550** is disposed in the second area B.

(87) [Disposition Structure of Components of Portable Air Purifier]

(88) Referring to FIGS. **1** to **4**, the accommodation space in the case **520** forming the skeleton of the portable air purifier **50** is divided into a first area A in the upper portion of the case **520** and a second area B in the lower portion of the case **520**.

(89) Components in relation to the suction, purification and discharge of air are disposed in the first area A. That is, the filter module **550** and the fan module **540**, **545** are disposed in the first area A, and accordingly, air flows in the first area A. A plurality of first inlets **570a** is provided as a passage for suctioning air, on the rear surface panel **570**. An air discharge part **532**, **533** and an outlet **510a** are provided as a passage for discharging air purified in the first area A, on the fan cover **530** and the front surface panel **510**. Additionally, a flow path connecting among the first inlet **570a**, the air discharge part **532**, **533** and the outlet **510a** is formed in the first area A.

(90) That is, the first inlet **570a**, the filter module **550**, the fan module **540**, **545**, the air discharge part **532**, **533** and the outlet **510a** are provided in the first area A, and a flow path for allowing air suctioned into the portable air purifier **50** to pass through the air purifier **50** is formed in the first area A.

(91) Components that do not directly relate to an air flow for air purification are disposed in the second area B. That is, the main PCB **595**, the battery **560** and the sensor module **600** are disposed in the second area B. The rear cover **580** covers the open rear of the space where the above components are disposed.

(92) In the embodiment, the case **520** is formed in to a cuboid having a length in the up-down direction greater than a length in the lateral direction. Additionally, the up-to-down length of the first area A in the upper portion of the case **520** is greater than that of the second area B in the lower portion of the case **520**. That is, when the portable air purifier **50** stands vertically, the first area A in the upper portion of the case **520** occupies more space than the second area B in the lower portion of the case **520**.

(93) A lower cover part **535** of the fan cover **530** is disposed at the frontmost side of the second area B. The battery **560** is disposed behind the lower cover part **535** while being disposed in the second area B. The fan module **540**, **545** and the filter module **550** are disposed over the battery **560**, and the rear surface cover **580** is disposed behind the battery **560**. Additionally, the sensor module **600** may be disposed between the battery **560** and the rear surface cover **580**.

(94) That is, the upper boundary of the battery **560** is defined by the fan module **540**, **545** and the filter module **550**, the lateral and lower boundaries of the battery **560** are defined by the first surface part **521** and the third surface part **527** of the case **520**, and the rear boundary of the battery **560** is accommodated in space defined by the rear surface cover **580**.

(95) In the embodiment, the battery **560** is a heavier object than the fan module **540**, **545** and the filter module **550**. It is preferable that the battery **560** weights more than a total weigh of the fan module **540**, **545** and the filter module **550**.

(96) Ordinarily, since the weight per unit volume of the battery 560 is much greater than that of the fan module 540, 545 and that of the filter module 550, the battery 560 may be readily provided as a heavier object than the fan module 540, 545 and the filter module 550 although the weight or size of the battery 560 does not increase intentionally.

(97) That is, even if a battery 560 having capacity required for the ordinary use of an portable air purifier 50 is applied to the portable air purifier 50, naturally, the battery 560 weights more than the fan module 540, 545 and the filter module 550.

(98) When the battery 560 as a heavy object is disposed in the lower portion of the portable air purifier 50, the following effects can be produced.

(99) First, when the battery 560 that is a heavy object is disposed in the lower portion of the portable air purifier 50, the center of gravity of the portable air purifier 50 is biased toward the lower side of the portable air purifier 50 from the up-to-down center of the portable air purifier 50. That is, the center of gravity of the portable air purifier 50 is biased toward the lower side of the portable air purifier 50, in which the battery 560 is disposed.

(100) The bias of the center of gravity of the portable air purifier 50 toward the lower portion of the portable air purifier 50, where the battery 560 is disposed, reduces the risk of the overturning of the portable air purifier 50, when the portable air purifier 50 stands vertically.

(101) That is, when the portable air purifier 50 stands vertically, the portable air purifier 50 rarely falls since the center of gravity of the portable air purifier 50 is at the lower side of the portable air purifier 50 because of the battery 560 in the lower portion of the portable air purifier 50.

(102) When it comes to the portable air purifier 50 in the embodiment, the third surface part 527 of the case 520 forms a flat surface having a greater surface area than the second surface part 523 of the case 520. That is, the third surface part 527 forms a long flat surface that has a greater length in the lateral direction than the second surface part 523. That is, under the assumption that the length of the third surface part 527 in the lateral direction is d1 and that the length of the upper surface part 527 in the lateral direction is d2, the relationship $d1 > d2$ is satisfied. Additionally, the second connection surface part 529 connecting between the third surface part 527 and the first surface part 521 forms a curved surface that has a less R value than the first connection surface part 525.

(103) The third surface part 527 of the case 520 is a portion contacting the bottom surface of the portable air purifier 50 when the portable air purifier 50 stands vertically. That is, the third surface part 527 is a portion that supports the portable air purifier 50 such that the portable air purifier 50 keeps sanding.

(104) Since the third surface part 527 of the case 520 forms a surface area that has a greater surface area than the second surface part 523, the third surface part 527 of the case 520 can support the portable air purifier 50 that stands vertically, more reliably.

(105) In combination of the structure in which the battery 560 is disposed in the lower portion of the portable air purifier 50 such that the center of gravity of the portable air purifier 50 is at the lower side of the portable air purifier 50, and the structure in which the third surface part 527 of the case 520, which supports the portable air purifier 50 standing vertically, forms a flat surface that has a greater surface area than the second surface area 523, effectively suppressing the overtraining of the portable air purifier 50 and reliably keeping the portable air purifier 50 standing vertically.

(106) Second, when the battery 560 as a heavy object is disposed in the lower portion of the portable air purifier 50, the other components such as the filter module 550 and the fan module 540, 545 constituting the portable air purifier 50 need to be disposed further upward than the battery 560. That is, components in relation to the suction, purification and discharge of air need to be disposed further upward than the battery 560.

(107) To ensure the charge capacity of the battery 560, required for the smooth operation of the portable air purifier 50, the battery 560 needs to have a predetermined size or greater. It means that the portable air purifier 50 needs to have installation space of a predetermined size or greater, therein, to install the battery 560. Additionally, it is irrational to form a flow path for allowing an

air to flow in the space where the battery **560** is installed. Accordingly, components in relation to the suction, purification and discharge of air needs to be disposed to avoid the battery **560**, i.e., in a position higher than the position of the battery **560**.

(108) In the disposition structure, a flow path for the suction, purification and discharge of air is formed in the first area A higher than the position of the battery **560**, in the portable air purifier **50**. Thus, the suction of air into the portable air purifier **50**, and the discharge of air purified in the portable air purifier **50** are performed in a position higher than the position of the battery **560**.

(109) As purified air is discharged from the upper portion of the portable air purifier **50**, the air purified in the portable air purifier **50** can reach the user's face more easily.

(110) When the portable air purifier **50** is placed and used on the bottom surface lower than the user's face, the portable air purifier **50** that stands vertically allows a greater amount of air purified in the portable air purifier **50** to reach the user's face than the portable air purifier **50** that lies horizontally.

(111) To this end, when purified air is discharged from the upper portion of the portable air purifier **50** in the state in which the portable air purifier **50** stands vertically, a greater amount of the air purified in the portable air purifier **50** reaches the user's face.

(112) In the embodiment, since the battery **560** is disposed in the lower portion of the portable air purifier **50**, a flow path for the suction, purification and discharge of air is formed in a position higher than the position of the battery **560**, in the portable air purifier **50**. Accordingly, purified air is discharged from the upper portion of the portable air purifier **50**, and a greater amount of the air purified in the portable air purifier **50** reaches the user's face.

(113) That is, in the structure where the battery **560** is disposed in the lower portion of the portable air purifier **50**, the structural stability of the portable air purifier **50** can improve to reduce the risk of overturning of the portable air purifier **50** that stands vertically, and an efficient flow path can also be formed to enable a greater amount of air purified in the portable air purifier **50** to reach the user's face.

(114) Third, the structure, in which the battery **560** as a heavy object is disposed in the lower portion of the portable air purifier **50** such that components in relation to the suction, purification and discharge of air are disposed in a position higher than the position of the battery **560**, may help to expand the range in which the portable air purifier **50** is installed.

(115) In an example, when the portable air purifier **50** is used in the state of being held in a cup holder h in a vehicle, the area where air is suctioned and the area where purified air is discharged are disposed higher than the cup holder, such that the portable air purifier **50** is reliably held in vehicle while maintaining a high level of air purification performance. To this end, the up-to-down length of the second area B where the battery **560** is disposed is set to the depth of the cup holder or greater, for example.

(116) In another example, if the lower area of the portable air purifier **50** is fixed by a tong type holder and the like, the portable air purifier **50** can be fixed stably while the portable air purifier **50**'s areas where air is suctioned and discharged are not blocked.

(117) That is, components such as a battery **560** that does not directly relate to an air flow for air purification are disposed in the lower portion of the portable air purifier **50**, and the portable air purifier **50** is held and fixed through its lower portion, ensuring a high level of air purification performance and a reliable fixation of the portable air purifier **50**.

(118) [Disposition Structure of Fan Module and Filter Module]

(119) The portable air purifier **50** in the embodiment may include a fan module **540**, **545** that suction air from the rear of the portable air purifier **50** and discharges air through the front of the portable air purifier **50**.

(120) The fan module **540**, **545** may be disposed between the fan cover **530** and the filter module **550**. That is, the fan cover **530** may be disposed at the front of the fan module **540**, **545**, and the filter module **550** may be disposed at the rear of the fan module **540**, **545**.

(121) While the fan module **540**, **545** is accommodated in the accommodation space of the case **520**, the fan module **540**, **545** may be disposed in the first area A. Accordingly, the fan module **540**, **545** may be disposed in a position that faces the rear surface of the upper cover part **531**, out of the upper cover part **531** of the fan cover **530** and the lower cover part **535** of the fan cover **530**.

(122) In the embodiment, the portable air purifier **50** is provided with two fan modules **540**, **545**, for example. At this time, the portable air purifier **50** is provided with a first fan module **540** and a second fan module **545**.

(123) The first fan module **540** may be disposed between the discharge surface and the filter module **550**, specifically, between the fan cover **530** and the filter module **550**. At this time, the first fan module **540** may be disposed in parallel with the suction surface and the discharge surface. The second fan module **545** and the first fan module **540** may be disposed on the same surface. That is, the second fan module **545** may be disposed in parallel with the suction surface and the discharge surface, between the fan cover **530** and the filter module **550**.

(124) The positions of the first fan module **540** and the second fan module **545** are disposed not to overlap each other in the third direction. To this end, the first fan module **540** and the second fan module **545** may be disposed in the third direction, i.e., in the up-down direction. That is, the first fan module **540** and the second fan module **545** may be disposed on the same perpendicular line.

(125) The first fan module **540** and the second fan module **545** may be disposed to face the air discharge part **532**, **533** formed on the fan cover **530**. Accordingly, air, suctioned into the first fan module **540** and the second fan module **545** through the rear surface panel **570**, may pass through an upper air discharge part **532** and a lower air discharge part **533** respectively, and then be discharge forward through the front surface panel **510**.

(126) In the embodiment, the first fan module **540** and the second fan module **545** may have the same size and shape, for example. However, the first fan module **540** and the second fan module **545** may have a different size and shape.

(127) The filter module **550** may be disposed between the suction surface and the fan module **540**, **545**. The filter module **550** includes a filter **559**, and the filter **559** filters air suctioned through the suction surface.

(128) The filter **559** may form a filter surface that forms a flat surface in a direction orthogonal to the directions in which the fan suction and discharges air. That is, the filter **559** may form a filter surface parallel with the suction surface and the discharge surface. The suction surface, the filter surface of the filter **559** and the discharge surface may be disposed in a straight line.

(129) Additionally, the filter **559** may form a filter surface parallel with the fan module **540**, **545**. The suction surface, the filter surface, the fan module **540**, **545** and the discharge surface may be disposed in a straight line.

(130) The positions of the first fan module **540** and the second fan module **545** in the first direction and the second direction may correspond to the area occupied by the filter surface. Further, the surface areas taken up by the first fan module **540** and the second fan module **545** may correspond to the surface area of the filter surface. For example, when viewed from the front, the first fan module **540** and the second fan module **545** may be disposed to overlap the filter surface.

(131) Furthermore, the directions of the first fan module **540** and the second fan module **545**'s suction of air may be the same as the directions of the suction surface and the discharge surface's disposition. That is, the first fan module **540** and the second fan module **545** may suction air respectively in the first direction or the axial direction.

(132) The surface area occupied by the fan module **540**, **545** including the first fan module **540** and the second fan module **545** described above may correspond to the surface area of the suction surface and the surface area of the filter surface. Additionally, the surface area taken by the fan module **540**, **545** may correspond to the surface area of the discharge surface. Under the assumption that the suction surface, the filter surface and the discharge surface have a corresponding surface area, the surface area occupied by the fan module **540**, **545** may correspond

to the surface area of each of the suction surface, filter surface and discharge surface.

(133) Accordingly, air can be suctioned and discharged by the fan module **540, 545** through the entire suction surface and the entire discharge surface.

(134) For example, air can be suctioned by the fan module **540, 545** through the suction surface that is ensured as much surface area as the fan module **540, 545**. Air suctioned through the suction surface can be filtered by the filter surface that is ensured as much surface area as the suction surface and the fan module **540, 545**. Additionally, air can be discharged by the fan module **540, 545** through the discharge surface that is ensured as much surface area as the fan module **540, 545**.

(135) That is, the suction surface and the discharge surface may ensure a passage having an optimal surface area required for the fan module **540, 545** to suction and discharge air. Additionally, the filter **559** may ensure a filter surface having an optimal surface area required to filter air suctioned through the suction surface. Thus, air may flow effectively, based on the operation of the fan module **540, 545**.

(136) In the embodiment, the suction surface, the filter surface, the fan module **540, 545** and the discharge surface are disposed in parallel. Additionally, the suction surface, the filter surface, the fan module **540, 545** and the discharge surface are disposed in the first direction, and air flows in the same direction. That is, air, flowing based on the operation of the fan module **540, 545**, can flow, in the same straight line, in the same direction as the direction in which the suction surface, the filter surface, the fan module **540, 545** and the discharge surface are disposed.

(137) As air flows in a straight line, resistance against the flow of the air decreases, and the air can flow more smoothly. Accordingly, a sufficient amount of air is suctioned, and in response, a sufficient amount of air is discharged by the fan module **540, 545**, leading to improvement in the air purification performance of the portable air purifier **50**.

(138) [Structure of Fan Module]

(139) As described above, the portable air purifier **50** in the embodiment may include a plurality of fan modules **540, 545**. In the embodiment, the portable air purifier **50** includes the first fan module **540** and the second fan module **545**, and the first fan module **540** and the second fan module **545** have the same structure, for example. Herein, the structure of the first fan module **540** is described as an example.

(140) FIG. **7** is a cross-sectional view specifically showing the structure of portion “VII” in FIG. **4**, FIG. **8** is a perspective view separately showing a fan module of one embodiment, and FIG. **9** is an perspective view separately showing a fan case in FIG. **8**. FIG. **10** is a front view separately showing a fan in FIG. **8**, FIG. **11** is a side view showing the fan in FIG. **10**, and FIG. **12** is a cross-sectional view along line “X II-X II” in FIG. **11**.

(141) In FIG. **7**, some components of the portable air purifier are omitted.

(142) Referring to FIGS. **3, 4, 7** and **8**, the first fan module **540** may include a shaft **5410**, a motor **5420**, a fan case **5430** and a fan **5440**.

(143) The shaft **5410** may extend in the first direction, i.e., the axial direction. One side of the shaft **5410** in the axial direction may connect to the fan **5450**. The other side of the shaft **5410** in the axial direction may connect to the fan case **5430**. Detailed description in relation to this is provided hereafter.

(144) The motor **5420** connects to the fan **5450** and applies a rotational force to the fan **5450**. For example, the motor **5420** may be provided in the form of a BLDC motor in which a frequency can be adjusted. The motor **5420** may include a stator **5421** and a rotor **5423**.

(145) The stator **5421** may be disposed at the center side of the motor **5420** in the radial direction of the motor **5420**, and the rotor **5423** may be disposed outside the stator **5421** in the radial direction of the stator **5421**. That is, the motor **5420** in the embodiment may be provided in the form of an outer motor.

(146) The stator **5421** may be fixed to the fan case **5430, 5440**. The rotor **5423** may be fixed to the fan **5450**. The rotor **5423** may rotate around the shaft **5410**, outside the stator **5421**, and the fan

5450 may rotate together with the rotor **5423** as the rotor **5423** rotates.

(147) The fan case **5430, 5440** may support the stator **5421** and the shaft. The fan case **5430, 5440** may include a first support part **5430** and a second support part **5440**, as illustrated in FIGS. 7 to 9.

(148) The first support part **5430** may be disposed at the center side of the fan case **5430, 5440** in the radial direction of the fan case **5430, 5440**. The first support part **5430** may include a support plate **5431** and a boss **5433**.

(149) The support plate **5431** may be formed into a circular plate. The support plate **5431** may form a flat surface that is parallel with at least any one of the suction surface, the filter surface and the discharge surface. The boss **5433** may protrude in the axial direction toward the fan **5450** from the support plate **5431**. In the embodiment, the boss **5433** is disposed at the center of the support plate **5431** in the radial direction of the support plate **5431**, for example.

(150) The stator **5421** may be installed in the first support part **5430**. The stator **5421** may be coupled to the boss **5433** while surrounding the boss **5433** from the outside in the radial direction. That is, the stator **5421** and the boss **5433** may be coupled in a way that the boss **5433** is fitted into the stator **5421** in the axial direction. The support plate **5431** may support the boss **5433** coupled to the stator **5421** at one side of the boss **5433** in the lateral direction, as described above.

(151) Additionally, one side of the stator **5421** coupled to the boss **5433** in the axial direction of the stator **5421** may face the support plate **5431**. While one side of the stator **5421** in the axial direction of the stator **5421** contacts the support plate **5431**, the stator **5421** and the support plate **5431** may be coupled. Certainly, the stator **5421** may be installed in the first support part **5430** in a way that the stator **5421** is spaced from the support plate **5431**.

(152) Further, the boss **5433** may have a hollow hole, therein. The shaft **5410** may be inserted into the boss **5433** through the hollow hole. The shaft **5410** may pass through the boss **5433** in the axial direction and protrude from the boss **5433**, and connect to a below-described hub **5451** of the fan **5450**, outside the boss **5433**.

(153) A bearing **5435** may be inserted into the boss **5433** having the hollow hole. The shaft **5410** may be coupled to the bearing **5435**, in the boss **5433**. Accordingly, the shaft **5410** may be rotatably supported by the bearing **5435**. In the embodiment, a pair of bearings **5435** is spaced a predetermined distance apart from each other in the axial direction, for example. The bearing **5435**, disposed as described above, may help to install the shaft **5410** rotatably in the boss **5433** in a more reliable manner.

(154) The second support part **5440** may support the first support part **5430**, and couple the first fan module **540** to at least any one of the fan cover **530** and the filter module **550**.

(155) The second support part **5440** may be disposed outside the first support part **5430** in the radial direction of the first support part **5430**. The second support part **5440** may support the first support part while surrounding the first support part **5430** from the outside in the radial direction.

(156) The second support part **5440** may be formed into an approximate rectangle. For example, the length of the second support part **5440** in the second direction may be determined to correspond to the length of the filter module **550** in the second direction.

(157) Additionally, the length of the second support part **5440** in the third direction may be determined to correspond to half the length of the filter module **550** in the third direction. That is, if a pair of second support parts **5440** is disposed in the third direction, the exterior shapes of the pair of second support parts **5440** and the filter module **550** may be approximately aligned, when viewed from the front.

(158) The second support part **5440** may have a hollow hole, therein. The hollow hole of the second support part **5440** may be formed in a way that the hollow hole penetrates the inside of the second support part **5440** in the axial direction. The hollow hole may have a radius greater than a radius of the support plate **5431**. That is, a hollow hole bigger than the support plate **5431** is formed in the second support part **5440**, and the support plate **5431** may be disposed in the second support part **5440**.

(159) Accordingly, the second support part **5440** may be spaced a predetermined distance apart from the first support part **5430**, specifically, from the support plate **5431** in the centrifugal direction.

(160) A fan outlet **5430a** may be formed between the support plate **5431** and the second support part **5440** that are spaced as described above. The fan outlet **5430a** may form a passage allowing air having flown into the first fan module **540** to pass through the first fan module **540** and escape from the first fan module **540**.

(161) The fan case **5430**, **5440** may further include a connection part **5445**. The connection part **5445** may extend from the support plate **5431** in the centrifugal direction and connect to the second support part **5440**. The connection part **5445** may connect between the support plate **5431** and the second support part **5440** and help the second support part **5440** to support the support plate **5431**.

(162) In the embodiment, a plurality of connection parts **5445** may be disposed between the support plate **5431** and the second support part **5440**. Additionally, each of the connection parts **5445** may have a width less than the length of the connection part **5445** in the radial direction. Each of the connection parts **5445**, formed as described above, may be spaced at a predetermined interval along the circumferential direction of the support plate **5431**.

(163) Accordingly, while the fan outlet **5430a** is formed between the support plate **5431** and the second support part **5440**, the fan outlet **5430a** may be respectively formed among the plurality of connection parts **5445**.

(164) Additionally, the connection part **5445** may provide a passage through which an electric wire, connected to the stator **5421** on the support plate **5431**, passes. The electric wire connecting to the stator **5421** may be withdrawn toward the second support part **5440** side through the area where the connection part **5445** is disposed. At this time, the electric wire may pass through the fan outlet **5430a** area in the state being covered by the connection part **5445**, and connect between the stator **5421** and the main PCB **595** (see FIG. 3). When viewed from the front, the electric wire that is withdrawn from the stator **5421** and passes through the fan outlet **5430a** area is cover by the connection part **5445** and is not be seen.

(165) The second support part **5440** may be provided with a side rib **5443**. The side rib **5443** may be disposed at at least any one of both sides of the second support part **5440** in the second direction of the second support part **5440**. The side rib **5443** may protrude from a lateral portion of the second support part **5440** in the second direction of the second support part **5440**.

(166) The length of the side rib **5443** in the second direction may be set to correspond to the length of the second support part **5440** in the second direction. Additionally, the length of the side rib **5443** in the first direction may be less than the length of the first support part **5430** in the first direction. That is, the side rib **5443** may have a thickness less than that of the first support part **5430**. For example, the side rib **5443** may have a thickness that is one fourth or less of the thickness of the first support part **5430**.

(167) The side rib **5443** may be biased toward the front or the rear of the second support part **5440** while protruding from the second support part **5440**. In the embodiment, the side rib **5443** is biased toward the rear of the second support part **5440**, for example.

(168) As the fan module **540**, **545** is inserted into the case **520**, the side rib **5443** may contact both lateral surfaces of the case **520**, i.e., the inner surfaces of the first surface parts **521**. Accordingly, predetermined space may be formed between a lateral surface of the second support part **5440** and the first surface part **521**.

(169) The formed space may be used as space for allowing an electric wire to pass. For example, an electric wire connecting to the sub PCB **595** disposed on the fan module **540**, **545** may pass through the space and connect to the stator **5421** of the motor **5420**.

(170) Referring to FIGS. 7 and 10 to 12, the fan **5450** may include a hub **5451** and fan blade **5455**.

(171) The hub **5451** is disposed at the center of the fan **5450** in the radial direction of the fan **5450**, and rotates together with the rotor **5423** and the shaft. The hub **5451** may include a first fan module

540, 545 and a skirt part **5433**.

(172) The first fan module **540, 545** may be formed into a circular plate parallel with the support plate **5431**. The first fan modules **540, 545** may be disposed in parallel with the support plate **5431** with the boss **5433** between the first fan modules **540, 545**.

(173) The first fan module **540, 545** may be provided with a shaft coupling part **5452a**. The shaft coupling part **5452a** may be disposed at the center of the first fan module **540, 545** in the radial direction of the first fan module **540, 545**. The shaft coupling part **5452a** may protrude in the axial direction toward the boss **5433** from the first fan module **540, 545**.

(174) The shaft coupling part **5452a** may be coupled to the end portion of the shaft **5410** in the axial direction of the shaft **5410**. For example, the shaft coupling part **5452a** may be coupled to the shaft **5410** in a way that the shaft **5410** is fitted into the shaft coupling part **5452a**. The shaft **5410** may be fixed to the shaft coupling part **5452a** or rotatably coupled to the shaft coupling part **5452a**.

(175) The skirt part **5453** may protrude toward the support plate **5431** from the edge of the hub plate part **5452**. The skirt part **5453** may form a slant surface that inclines in the centrifugal direction as the skirt part **5452** becomes farther from the hub plate part **5452** in the axial direction. For example, the shape in which the hub plate part **5452** and the skirt part **5453** connect may be a truncated cone shape which has a hollow hole therein and one side of which is open. The skirt part **5433** may be disposed outside the stator **5421** in the radial direction of the stator **5421**. That is, the stator **5421** may be disposed in space surrounded by the skirt part **5453** and the hub plate part **5452**.

(176) The fan blade **5455** may protrude from the hub **5451** in the centrifugal direction. The fan **5450** may be provided with a plurality of fan blades **5455**, and the fan blades **5455** may be spaced a predetermined distance apart from one another along the circumference direction of the hub **5451**.

(177) Specifically, the fan blade **5455** may protrude from the skirt part **5453** in the centrifugal direction. At this time, the inside of the fan blade **5455** in the radial direction thereof may connect to the skirt part **5453**, and the outside of the fan blade **5455** in the radial direction thereof may connect to a shroud **5457** that is described hereafter. That is, the skirt part **5453** is a portion of the hub **5451**, which directly connects to the fan blade **5455** and directly contacts air passing through the first fan module **540**. The skirt part **5453** may closely relate to a flow path of air passing through the first fan module **540**.

(178) The fan **5450** may further include a shroud **5457**. The shroud **5457** may be spaced a predetermined distance apart from the hub **5451** in the radial direction, outside the hub **5451** in the radial direction thereof. The shroud **5457** may be spaced from the hub **5451** by a distance corresponding to the length of the fan blade **5455** in the radial direction of the fan blade **5455**. Additionally, each of the fan blades **5455** may connect between the hub **5451**, specifically, the skirt part **5453** and the shroud **5457**.

(179) The shroud **5457** may form a slant surface that inclines in the centripetal direction as the shroud **5457** becomes farther from the fan case **5430, 5440** in the axial direction, i.e., toward the rear. For example, the shroud **5457** may form a slant surface that is approximately parallel with the skirt part **5453**. In the embodiment, a gap between the skirt part **5453** and the shroud **5457** increases further toward the front, for example.

(180) Each of the fan blades **5455**, connecting between the shroud **5457** and the skirt part **5433**, may include a leading edge **5455a**, a trailing edge **5455b**, a shroud chord **5455c** and a hub chord **5455d**.

(181) The leading edge **5455a** may be disposed at the front end of the fan **5450** in a rotation direction and formed in a straight manner. The rotation direction is defined as a direction in which the fan **5450** rotates. The leading edge **5455a** may be formed into a straight line that is disposed at the front end of the fan **5450** in the rotation direction of the fan **5450** and extends in the radial direction.

(182) The trailing edge **5455b** may be disposed at the rear end of the fan in the rotation direction and formed in a straight manner. The trailing edge **5455b** may be formed into a straight line that

extends in a direction between the axial direction and the radial direction.

(183) The shroud chord **5455c** may connect one end of the leading edge **5455a** and one end of the trailing edge **5455b**. The shroud chord **5455c** may extend from the inner circumferential surface of the shroud **5457**.

(184) The hub chord **5455d** may connect the other end of the leading edge **5455a** and the other end of the trailing edge **5455b**. The hub chord **5455d** may extend from the outer circumferential surface of the hub **5451**.

(185) Additionally, one end of the leading edge **5455a** and one end of the trailing edge **5455b** may connect to the inner circumferential surface of the shroud **5457**. The other end of the leading edge **5455a** and the other end of the trailing edge **5455b** may connect to the outer circumferential surface of the skirt part **5453**.

(186) One end of the leading edge **5455a** may be disposed closer to the center of the hub plate part **5452** in the radial direction of the hub plate part **5452** than one end of the trailing edge **5455b**. The other end of the leading edge **5455a** may be disposed closer to the center of the hub plate part **5452** in the radial direction of the hub plate part **5452** than the other end of the trailing edge **5455b**, since one end and the other end of the leading edge **5455a** is disposed further forward than one end and the other end of the trailing edge **5455b** in the rotation direction, and the radius of the skirt part **5453** decreases further toward the front in the rotation direction.

(187) Additionally, the hub **5451** may include an inner projection part **5454**. The inner projection part **5454** may protrude toward the support plate **5431** from the hub plate part **5452**. In the embodiment, the inner projection part **5454** and the skirt part **5453** protrude from the same point on the hub plate part **5452**, e.g., from the edge of the hub plate part **5452**, for example.

(188) The inner projection part **5454** may be formed in a way that the inner projection part **5454** extends from the hub plate part **5452** in the axial direction. For example, the shape in which the hub plate part **5452** and the skirt part **5453** connect may be a cylinder shape which has a hollow hole therein and one side of which is open.

(189) The inner projection part **5454** may be disposed between the skirt part **5453** and the stator **5421**. That is, the stator **5421** may be disposed in space surrounded by the inner projection part **5454** and the hub plate part **5452**.

(190) The rotor **5423** may be disposed between the inner projection part **5454** and the stator **5421**. The rotor **5423** may be fixed onto the inner circumferential surface of the inner projection part **5454**. That is, the stator **5421** may be fixed to the fan case **5430**, **5440**, and the rotor **5423** may be fixed to the fan **5450**. Accordingly, the fan **5450** may rotate together with the rotor **5423** as the rotor **5423** rotates.

(191) In the embodiment, the fan blade **5455** may connect to the skirt part **5453** of the hub **5451**. To induce the flow of air flowing into the fan module **540**, **545** in a direction between the axial direction and the radial direction, the skirt part **5453** forms a slant surface that inclines in a direction between the axial direction and the radial direction.

(192) Since the skirt part **5453** is formed at a slant, it is difficult to fix the rotor **5423** to the inner circumferential surface of the skirt part **5453**. For example, for the rotor **5423** to be fixed to the inner circumferential surface of the skirt part **5453**, the rotor **5423** needs to be entirely formed into an approximate truncated cone having a hollow hole since the shape of the outer circumferential surface of the rotor **5423** needs to correspond to the shape of the inner circumferential surface of the skirt part **5453**. However, the shape of the rotor **5423** may be inappropriate for the reliable driving of the motor **5420**.

(193) Considering the fact, in the embodiment, the inner projection part **5454** is provided between the skirt part **5453** and the stator **5421**. The inner projection part **5454** may provide a fixation surface that allows the rotor **5423** to be fixed stably to the inside of the hub **5451** when the rotor **5423** is entirely formed into a cylinder having a hollow hole.

(194) Additionally, the inner projection part **5454** may form a structure for improving the rigidity of

the hub **5451**, in the hub **5451**. The inner projection part **5454** may help to improve the rigidity of the entire fan **5450** effectively while suppressing a significant increase in the weight of the entire fan **5450**.

(195) [Structure of Coupling Between Fan Module and Fan Cover]

(196) FIG. **13** is an exploded perspective view separately showing a fan cover and a fan module.

(197) Referring to FIG. **13**, the fan cover **530** may include an upper cover part **531** and a lower cover part **535**.

(198) The upper cover part **531** is disposed at the front of the fan module **540**, **545**. The upper cover part **531** may be provided with an air discharge part **532**, **533**. The air discharge part **532**, **533** may be formed in a way that a portion of the upper cover part **531** penetrates or is cut. The air discharge part **532**, **533** may form a passage that connects between the front of the case **520**, i.e., the discharge surface, and the fan **5450** of the fan module **540**, **545**, on the fan cover **530**.

(199) In the embodiment, the portable air purifier **50** is provided with two fan modules **540**, **545**, for example. That is, the first fan module **540** and the second fan module **545** are disposed vertically in the accommodation space of the case **520**.

(200) Accordingly, the fan cover **530** may also be provided with two air discharge parts **532**, **533**. That is, an upper air discharge part **532** and a lower air discharge part **533** may be disposed vertically at the upper cover part **531**.

(201) Air having passed through the first fan module **540** is discharged forward through the upper air discharge part **532**, and air having passed through the second fan module **545** may be discharged forward through the lower air discharge part **533**.

(202) In another example, the portable air purifier may be provided with one fan assembly or three or more fan assemblies, and accordingly, the fan cover may be provided with one air discharge part or three or more air discharge parts.

(203) Additionally, the upper cover part **531** may be provided with a first fastening projection **534**. The first fastening projection **534** may protrude from the rear surface of the upper cover part **531** rearward. In response, the fan case **5430**, **5440** may be provided with a fastening hole **5441**. The fastening hole **5441** may be formed in a way that penetrates in the front-rear direction in the second support part **5440**.

(204) In an example, a total of four fastening holes **5441** may be provided, and each of the fastening holes **5441** may be disposed at each edge of the second support part **5440**. The first fastening projection **534** may be disposed respectively in positions corresponding to the positions of the fastening holes **5441**.

(205) Each of the first fastening projections **534** may be inserted into the fastening hole **5441** and fitted into and coupled to the second support part **5440**. As a result of the coupling between the first fastening projection **534** and the second support part **5440**, the fan cover **530** and the fan module **540**, **545** may be coupled at a plurality of points.

(206) As the fan cover **530** and the fan module **540**, **545** are coupled, the fan module **540**, **545** may be fixed to the rear of the fan cover **530**. At this time, the fan module **540**, **545** may be fixed in the position where the fan outlet **5430a** formed at the front of the fan module **540**, **545** overlaps the air discharge part **532**, **533** formed at the fan cover **530** in the first direction. Thus, a straight line passage may be formed to allow air discharged from the fan module **540**, **545** to pass through the fan cover **530**.

(207) [Structure of Fan Base]

(208) FIG. **14** is an exploded perspective view separately showing a fan module, a fan base and a filter, FIG. **15** is a rear perspective view showing the rear surface of a fan base, and FIG. **16** is a perspective view showing a coupling state between the fan base and the filter.

(209) Referring to FIGS. **7** and **13** to **16**, the portable air purifier **50** according to the present disclosure may further include a fan base **5460**.

(210) The fan base **5460** may be disposed between the filter module **550** and the fan module **540**,

545. The fan base 5460 may be formed into a shape corresponding to the shape of the filter surface. For example, the fan base 5460 may be formed into a shape of the filter 559 viewed from the front, i.e., a rectangular shape. The fan base 5460 may include a base plate 5461, and a bell mouth 5463. (211) In this embodiment, two fan bases 5460 disposed in the third direction, i.e., the up-down direction, are disposed between the filter module 550 and the fan module 540, 545, for example. At this time, one fan base 5460 disposed in the upper portion is disposed between the first fan module 540 and the filter module 550, and the other fan base 5460 disposed in the lower portion is disposed between the second fan module 545 and the filter module 550.

(212) When the fan base 5460 is provided separately depending on the number of the fan modules, the fan base 5460 may be provided in response to the number of the fan modules even if the number of the fan modules varies. That is, when one fan module is provided, one fan base 5460 is applied, and when two or more fan modules are provided, the same number of the fan bases 5460 as the number of the fan modules may be stacked in the up-down direction. Thus, the fan base 5460 is provided regardless of the number of the fan modules, ensuring ease of management of components.

(213) Despite the fact, the fan base 5460 may be provided in a way that a single fan base 5460 includes a plurality of fan inlets 5462.

(214) The base plate 5461 may be disposed between the filter module 550 and the fan module 540, 545. The length of the base plate 5461 in the first direction may be much less than that of the filter module 550 and the fan module 540, 545. For example, the base plate 5461 may be formed into a rectangular plate.

(215) The base plate 5461 may have a fan inlet 5462. The fan inlet 5462 may be formed in a way that fan inlet 5462 penetrates on the base plate 5461 in the first direction. The fan inlet 5462 may be disposed approximately in the position where the fan inlet 5462 overlaps the air discharge part 532, 533 (see FIG. 13) of the fan cover 530 and the fan outlet 5430a of the fan module 540, 545 in the first direction. The fan inlet 5462 formed as described above may form a passage that connects between the filter 559 and the fan module 540, 545, at the fan base 5460.

(216) The bell mouth 5463 may protrude from the base plate 5461. The bell mouth 5463 may protrude from the base plate 5461 to the fan module 540, 545, in the axial direction.

(217) In the embodiment, the fan inlet 5462 has the same shape as the shroud 5457, i.e., a circular shape, for example. Additionally, the radius of the fan inlet 5462 may be set similarly to the radius of a portion of the shroud 5457, which is adjacent to the fan inlet 5462.

(218) The bell mouth 5463 may be formed in a way that the bell mouth 5463 surrounds the outer circumferential surface of the fan inlet 5462 formed as described above. In other words, the fan inlet 5462 may be formed in a way that the fan inlet 5462 penetrates at the inside of the bell mouth 5463 in the radial direction of the bell mouth 5463.

(219) The bell mouth 5463 may protrude toward the fan module 540, 545 from the fan inlet 5462 in the first direction. At this time, at least a portion of the bell mouth 5463 may be inserted into the shroud 5457 in the radial direction of the shroud 5457. The bell mouth 5463 guides a suction flow at the entrance of the fan module 540, 545, and help to improve the suction and discharge performance of the fan module 540, 545.

(220) The fan base 5460 may be coupled to the fan base 5460. To this end, any one of the fan base 5460 and the filter module 550 may have a fastening boss 5433, and any one of the fan base 5460 and the filter module 550 may have a second fastening projection 553. In the embodiment, the fan base 5460 has the fastening boss 5433, and the filter module 550 has the second fastening projection 553, for example.

(221) The fastening boss 5433 may protrude from the base plate 5461 to the filter module 550 in the first direction. The fastening boss 5433 may have a hollow hole. The second fastening projection 553 may protrude from the filter module 550, specifically, the front surface of the filter case 551, toward the fan base 5460 in the first direction. The second fastening projection 553 may

be fitted into and coupled to the fastening boss **5433** in a way that the second fastening projection **553** is inserted into the hollow hole of the fastening boss **5433**.

(222) In an example, a total of four fastening bosses **5433** are provided, and each of the fastening bosses **5433** may be disposed at each edge of the first support part **5430**. The second fastening projection **553** may be disposed respectively in positions corresponding to the positions of the fastening bosses **5433**.

(223) Based on the coupling between the second fastening projection **553** and the fastening boss **5433**, the fan base **5460** and the filter module **550** may be coupled at a plurality of points. As the fan base **5460** is coupled to the filter module **550** as described above, the fan base **5460** may be fixed to the front of the filter module **550**.

(224) Additionally, the fan base **5460** may be coupled to the fan module **540**, **545**. To this end, any one of the fan base **5460** and the fan module **540**, **545** is provided with a third fastening projection **5467**, and the other may be provided with a projection boss **5442**. In the embodiment, the fan base **5460** is provided with the third fastening projection **5467**, and the fan module **540**, **545** is provided with the projection boss **5442**, for example.

(225) The projection boss **5442** may protrude from the second support part **5440** toward the fan base **5460** in the first direction. The projection boss **5442** may have a hollow hole. The third fastening projection **5467** may protrude from the front surface of the base plate **5461** toward the front. The third fastening projection **5467** may be fitted into and coupled to the fastening boss **5433** in a way that the third fastening boss **5467** is inserted into the hollow hole of the projection boss **5442**.

(226) As the projection boss **5442** and the third fastening projection **5467** are coupled, the fan base **5460** and the fan module **540**, **545** may be coupled such that the second support part **5440** is spaced a predetermined distance apart from the base plate **5461**. At this time, the predetermined distance is set to the length of a portion of the fan **5450** in the axial direction, protruding outward in the radial direction of the second support part **5440**, e.g., the length of the shroud **5457** in the first direction, or greater.

(227) Further, a total of four projection bosses **5442** are provided, and each of the projection bosses **5442** may be disposed at each edge of the second support part **5440**. The third fastening projection **5467** may be disposed respectively in positions corresponding to the positions of the projection bosses **5442**.

(228) A gap between the portions where each of the projection bosses **5442** and each of the third fastening projections **5467** are coupled may be open in the radial direction. Additionally, a portion of the fan **5450**, specifically, a portion of the shroud **5457**, may protrude outward in the radial direction of the second support part **5440** through the open portions.

(229) The fan base **5460** described above may serve as a coupling medium for coupling between the fan module **540**, **545** and the filter module **550**, and guide a suction flow at the entrance of the fan module **540**, **545** to improve the suction and discharge performance of the fan module **540**, **545**.

(230) [Aspect of Air Flow of Portable Air Purifier]

(231) FIG. **17** is a view showing an aspect of the air flow of the portable air purifier of one embodiment.

(232) In FIG. **17**, some components are omitted from the portable air purifier.

(233) Hereafter, an aspect of the air flow of the portable air purifier in the embodiment is described with reference to FIGS. **4** and **17**.

(234) Referring to FIGS. **4** and **17**, as the fan module **540**, **545** operates, air behind the portable air purifier **50** flows into the portable air purifier **50**. At this time, the air behind the portable air purifier **50** may pass through the suction surface through the first suction inlet **570a** formed on the rear surface cover **580**.

(235) The air having passed through the suction surface and having flown into the portable air purifier **50** passes through the filter **559**, and while the air passes through the filter **559**, the filter **60**

can filter physical particles such as dust/fine dust/ultra fine dust and the like, chemical substances such as odorant particles/harmful gases and the like, and microorganisms such as germs/viruses and the like, that are included in the air.

(236) At this time, the air may be suctioned into the filter **559** through the suction surface that has as much surface area as the filter surface. Additionally, the air suctioned through the suction surface may be filtered through the filter surface that has as much surface area as the fan module **540, 545**.

(237) That is, air may be suctioned and filtered effectively through the suction surface and the filter surface that have sufficient surface areas corresponding to the surface area of the fan module **540, 545**. Additionally, since the suction surface, the filter surface and the fan module **540, 545** are disposed in a straight line, air may be suctioned and filtered effectively while flow loss is minimized.

(238) The air having passed through the filter **559**, i.e., purified air, may flow into the fan module **540, 545** through the fan inlet **5462**. The flow of the air passing through the fan inlet **5462** may be guided by the bell mouth **5463**, and accordingly, a smooth inflow of air to the fan module **540, 545** may be induced effectively.

(239) The air having flown into the fan module **540, 545** may be discharged in a mixed flow direction while being discharged from the front of the fan module **540, 545** through the fan outlet **5430a**. The mixed flow direction may be defined as a front-side diagonal direction.

(240) The air, discharged from the front of the fan module **540, 545**, i.e., purified air, passes through the fan cover **530**, and is discharged from the front of the portable air purifier **50**. At this time, the purified air may pass through the discharge surface through the discharge outlet **510a** formed on the front cover **510**.

(241) The purified air may be discharged through the discharge surface that has as much surface area as the suction surface, the filter surface and the fan module **540, 545**. That is, the purified air may be discharged effectively through the discharge surface that has a sufficient surface area corresponding to the surface area of the suction surface, the surface area of the filter surface and the surface area of the fan module **540, 545**. Further, since the suction surface, the filter surface, the fan module **540, 545** and the discharge surface are disposed in a straight line, air may be suctioned and filtered, and purified air may be discharged, effectively, while flow loss is minimized.

(242) [Effect of Portable Air Purifier with Mixed Flow Fan-Type Fan]

(243) The portable air purifier **50** in the embodiment include a mixed flow fan type-fan **5450**. The portable air purifier **50** may produce the following effects.

(244) First, since the fan module **540, 545** includes a mixed flow fan-type fan **5450**, the fan module **540, 545** may ensure improvement in air purification performance, compared to a fan module including an axial flow fan-type fan.

(245) The axial flow fan allows air to be suctioned and discharged in a straight line. The axial flow fan may have a small thickness, and can be applied to a small-sized air purifier.

(246) However, the performance of the axial flow fan significantly deteriorates at fixed pressure. For example, if the density or the thickness of the filter **559** increases to enhance filtering performance, the suction and discharge performance of the axial flow fan may significantly deteriorate. Accordingly, it is difficult to apply an axial flow fan to a portable air purifier with a high performance filter.

(247) The mixed flow fan is somewhere between an axial flow fan and a centrifugal fan, and ensures more excellent performance at fixed pressure than the axial flow fan on condition that the mixed flow fan has the same size as the axial flow fan. However, the length of a mixed flow fan in the axial direction is greater than that of an axial flow fan on condition that the mixed flow fan has the same size as the axial flow fan.

(248) However, the axial flow fan is inappropriate to send high-pressure high-speed air.

Accordingly, a portable air purifier to which an axial flow fan is applied needs to further include a structure such as a guide vane. For example, a guide vane may be formed in the air discharge part

532, 533 of the fan cover **530**. The guide vane may be a cause for an increase in the length of the fan cover **530** in the first direction, i.e., the thickness of the fan cover **530**.

(249) An increase in the thickness of the fan cover **530**, caused by the guide vane, results in a decrease in the length of the axial flow fan in the first direction due to the limited space in the portable air purifier, and a deterioration in the suction and discharge performance of the axial flow fan.

(250) Unlike an axial flow fan, a mixed flow fan is appropriate to send high-pressure high-speed air. That is, without a structure such as a guide vane, the mixed flow fan may send air much farther than the axial flow fan.

(251) For this reason, in the embodiment, the fan cover **530** excludes a structure such as a guide vane. Accordingly, the thickness of the fan cover **530** decreases, and the fan module **540, 545** is small enough to be inserted into the air discharge part **532, 533**. For example, the length of the fan module **540, 545** in the first direction may be determined to the extent that the front surface of the fan module **540, 545** and the front surface of the fan cover **530** constitute the same surface. That is, the length of the fan module **540, 545** in the first direction may extend by a length occupied by a guide vane.

(252) As the size of the fan module **540, 545** increases, the suction and discharge performance of the fan module **540, 545** improves, enabling the portable air purifier **50** in the embodiment to ensure improvement in air purification performance.

(253) Additionally, in the structure of the fan module **540, 545**, the fan case **5430, 5440** does not surround the fan **5450** from the outside in the radial direction. That is, the fan **5450** may be formed to protrude further in the centrifugal direction than the inner circumferential surface of the second support part **5440**. Accordingly, at least a portion of the fan **5450** may protrude to a gap between the second support part **5440** and the base plate **5461**.

(254) Since the fan **5450** protrudes further in the centrifugal direction than the inner circumferential surface of the second support part **5440** as described above, the size of the fan **5450** may increase.

(255) As the size of the fan module **540, 545** increases as described above, the suction and discharge performance of the fan module **545, 545** improves, enabling the portable air purifier **50** in the embodiment to ensure improvement in air purification performance.

(256) Second, since the fan module **540, 545** includes a mixed flow fan-type fan **5450**, the range in which the portable air purifier **50** discharges air may expand.

(257) A mixed flow fan can discharge air in the mixed flow direction, and accordingly, purified air discharged through the discharge surface may be discharged in the mixed flow direction, i.e., a direction between the front and the centrifugal direction.

(258) When it comes to a small-sized portable air purifier **50**, if purified air is discharged toward the front only in a straight line, the range of the discharge of the purified air is limited to a range corresponding to the discharge surface.

(259) If the purified air is discharged within a range corresponding to the discharge surface, even considering the properties of the small-sized portable air purifier **50**, limitations are imposed on the use of the portable air purifier **50**.

(260) For example, the user has to keep holding up the portable air purifier **50** or has to accurately adjust the position of the portable air purifier **50** such that the discharge outlet **510a** faces the user's face, causing inconvenience to the user.

(261) The portable air purifier **50** in the embodiment may discharge high-pressure high-speed air in a direction between the front and the centrifugal direction. Accordingly, purified air may spread to a large area as well as reaching a far corner, enabling the portable air purifier **50** to ensure improvement in air purification performance.

(262) Third, each fan module **540, 545** includes the mixed flow fan-type fan **5450**, and a plurality of fan modules **540, 545** are stacked, allowing purified air to be discharged far away.

(263) As described above, a mixed flow fan can discharge air in the mixed flow direction.

(264) At this time, a collision between air discharged from the first fan module **540** and air discharged from the second fan module **545** may occur, in the portion where the fan modules **540**, **545** are adjacent to each other.

(265) For example, air may be discharged downward at a slant at the lower side of the first fan module **540** adjacent to the second fan module **545**, and air may be discharged upward at a slant at the upper side of the second fan module **545** adjacent to the first fan module **540**.

(266) At this time, a portion of the air discharged from the first fan module **540** and a portion of the air discharged from the second fan module **545** may collide with each other. Thus, their radiuswise speed vector decreases, and their frontwise speed vector may increase.

(267) A decrease in the radiuswise speed vector and an increase in the frontwise speed vector denote the direction of speed vector of air discharged from the portable air purifier **50** changes to a direction close to the front.

(268) As a result, the portable air purifier **50** discharges air mainly toward the front, and allows the air to reach a far corner, ensuring improvement in air purification performance.

(269) [Another Example of Fan Base]

(270) FIG. **18** is a view showing the structure of a fan base provided in a portable air purifier of another embodiment, and FIG. **19** is a view showing the structure of a fan base provided in the portable air purifier of one embodiment. FIG. **20** is a cross-sectional view showing the structures of the fan and the fan base in FIG. **19**, and FIG. **21** is a cross-sectional view showing the structures of the fan and the fan base in FIG. **18**.

(271) Hereafter, the structure of the portable air purifier of another embodiment is described with reference to FIGS. **18** to **21**.

(272) Referring to FIG. **18**, a fan base **6460** of the portable air purifier in another embodiment may include a base plate **5461**, a bell mouth **6463**, and a projection rib **6468**.

(273) As described with reference to the above embodiment, the base plate **5461** may be formed into a rectangular plate, and a fan inlet **5462** may be formed at the base plate **5461** in a way that penetrates in the first direction.

(274) The bell mouth **6463** may protrude from the inner circumferential surface of the base plate **5461** in the axial direction. A surface of the bell mouth **6463**, facing the filter module **550**, is referred to as a first surface **6463a** of the bell mouth **6463**, and a surface of the bell mouth **6463**, facing the fan module **540**, **545** (see FIG. **4**) is referred to as a second surface **6463b** of the bell mouth **6463**. Accordingly, the first surface **6463a** of the bell mouth **6463** is convex toward the filter module **550**, and the second surface **6463b** is concave.

(275) For example, the first surface **6463a** of the bell mouth **6463** is convex toward the filter module **550** in a rounded manner, and the second surface **6463b** of the bell mouth **6463**, which is the opposite surface of the first surface, may be depressed in a way that corresponds to the shape of the first surface of the bell mouth **6463**. Accordingly, space surrounded by the second surface **6463b** of the bell mouth **6463** may be formed in the bell mouth **6463**.

(276) At least a portion of the fan **5450** may be inserted into the space surrounded by the second surface **6463b** of the bell mouth **6463**. Thus, at least a portion of the bell mouth **6463** may be disposed to overlap the fan module **540**, **545** (see FIG. **4**).

(277) Specifically, at least a portion of the shroud **5457** may be inserted into the space surrounded by the second surface **6463b** of the bell mouth **6463**. In the embodiment, the bell mouth **6463** is formed into a half ring, for example. At this time, the bell mouth **6463** may be entirely formed into a half ring, and the longitudinal cross section of the bell mouth **6463** may have a half ring shape.

(278) Accordingly, space having a half-circle cross section may be formed in the bell mouth **6463**. At least a portion of the shroud **5457** may be inserted into the space having a half-circle cross section.

(279) The shroud **5457** may be disposed closest to the bell mouth **6463**, at the fan **5450**. The shroud **5457** may be provided with a shroud projection **5458**, at the end portion in the first direction,

specifically, the end portion of the rear side of the shroud **5457**.

(280) The shroud projection **5458** may protrude rearward from the end portion of the rear side of the slant surface of the shroud **5457**. The shroud projection **5458** may form a protruding surface that extends in the first direction, i.e., in a parallel direction with the direction in which the fan module **540**, **545** and the filter module **550** (see FIG. 7) are disposed, unlike the slant surface of the shroud **5457**.

(281) At least a portion of the shroud projection **5458** may be inserted into the space surrounded by the second surface **6463b** of the bell mouth **6463**. Accordingly, at least a portion of the fan **5450** may be inserted into the space surrounded by the second surface **6463b** of the bell mouth **6463**, and at least a portion of the bell mouth **6463** may be disposed to overlap the fan module **540**, **545**.

(282) The shroud projection **5458**, provided as described above, may block a gap in the lateral direction between the fan **5450** and the fan base **6460**, together with the bell mouth **6463**, and suppress generation of whirlwind between fan base **6460** and the fan **5450**.

(283) Additionally, the shroud projection **5458** may result in an increase in the thickness of the inner circumferential surface of the shroud **5457**, improving the rigidity of the fan **5450**.

(284) The fan base **6460** may further include a projection rib **6468**. The projection rib **6468** may protrude from the base plate **5461** and be disposed outside the bell mouth **6463** in the radial direction of the bell mouth **6463**. In the embodiment, the projection rib **6468** protrudes toward the shroud **5457** from the base plate **5461** in the axial direction while surrounding the bell mouth **6463** from the outside in the radial direction, for example.

(285) The projection rib **6468** may protrude in a way the forms a slant surface. The slant surface of the projection rib **6468** may be formed into a slant surface that is spaced a predetermined distance apart from the shroud **5457** and is parallel with the slant surface of the shroud **5457**.

(286) The projection rib **6468** may form a slant surface that extends in the centrifugal direction from the end portion outside the bell mouth **6463** in the radial direction of the bell mouth **6463**. At this time, the slant surface of the projection rib **6468** may be formed into a slant surface that inclines forward toward the centrifugal direction. At this time, the gradient of the slant surface of the projection rib **6468** may be the same as the gradient of the slant surface of the shroud **5457**. Accordingly, the slant surface of the projection rib **6468** may be spaced a predetermined distance apart from the shroud **5457**, and formed into a slant surface parallel with the slant surface of the shroud **5457**.

(287) Further, the projection rib **6468** may be integrated with the base plate **5461**. Specifically, the base plate **5461**, the bell mouth **6463** and the projection rib **6468** may be integrally formed.

(288) [Operation and Effect of Fan Base with Projection Rib]

(289) FIG. 19 is a view showing the structure of a fan base without a projection rib, FIG. 20 is a cross-sectional view showing the structures of the fan and the fan base in FIG. 19, and FIG. 21 is a cross-sectional view showing the structures of the fan and the fan base in FIG. 18.

(290) Referring to FIGS. 19 and 20, a gap of a predetermined distance may be formed between the fan base **5460** and the fan **5450**. Specifically, a gap of a predetermined distance may be formed between the base plate **5461** and the bell mouth **5463** of the fan base **5460** and the shroud **5457** of the fan **5450**.

(291) The fan base **5460** is fixed to the filter module **550** (see FIG. 7) and the fan case **5440**, and the fan **5450** is rotatably provided. Accordingly, a gap of a predetermined distance needs to be formed between the fan base **5460** and the fan **5450**.

(292) Considering the shape of the fan base **5460** and the shape of the fan **5450**, the narrowest gap between the fan base **5460** and the fan **5450** is formed at the point where the boundary portion between the bell mouth **5463** and the base plate **5461** faces the shroud **5457** in the axial direction (hereafter, a “minimum gap point”).

(293) At the point, a portion of the bell mouth **5463**, disposed at a frontmost side, is placed. Additionally, a portion of the shroud, disposed relatively rearward, is placed at the point. Certainly,

the rear most end portion of the shroud **5457** is not place at the point. However, a gap between the bell mouth **5463** and the shroud **5457** at a point further rearward than the minimum gap point P1 may be greater than at the minimum gap point P1, because of the half ring shape of the bell mouth **5463**.

(294) The gap between the fan base **5460** and the fan **5450** may increase toward the centrifugal direction from the minimum gap point P1. That is, a gap that increases toward the centrifugal direction is formed between the fan base **5460** and the fan **5450**.

(295) In relation to pressure distribution in the fan module **540**, **545** (see FIG. 4), pressure is relatively high at the discharge side of the fan **5450**, i.e., the outlet side **5430a** (see FIG. 7), and pressure is relatively low at the suction side of the fan **5450**, i.e., near the fan base **5460**.

Accordingly, whirlwind may be generated between the fan base **5460** and the fan **5450**, and may flow into the suction side of the fan **5450** through the gap between the fan base **5460** and the fan **5450**, specifically, the gap between the bell mouth **5463** and the shroud **5457**.

(296) In the above structure, a gap that increases toward the centrifugal direction is formed between the fan base **5460** and the fan **5450**. That is, a wide entrance may be formed between the fan base **5460** and the fan **5450**, making it easier for whirlwind to flow into the suction side of the fan through the gap between the bell mouth **5463** and the shroud **5457**.

(297) The whirlwind flowing into the suction side of the fan **5450** may interfere with the flow of air, flowing into the fan **5450** through the fan inlet **5462**, causing flow loss.

(298) To reduce the effect of the whirlwind, the gap between the bell mouth **5463** and the shroud **5457** may decrease. As the gap between the bell mouth **5463** and the shroud **5457** becomes narrower, the possibility that whirlwind is generated and flows into the fan **5450** may be reduced. Thus, a decrease in the gap between the bell mouth **5463** and the shroud **5457** may lead to a reduction in flow loss.

(299) However, a narrow gap between the bell mouth **5463** and the shroud **5457** is highly likely to cause contact between the fan **5450** and the bell mouth **5463**. For example, as the fan **5450** moves rearward due to an external force applied to the portable air purifier, the fan **5450** and the bell mouth **5463** may contact. That is, a narrower gap between the bell mouth **5463** and the shroud **5457** is highly likely to result in contact between the fan **5450** and the bell mouth **5463**, which is caused by an external force.

(300) The portable air purifier that is usually carried for use is inherently affected by an external force more frequently than another type of air purifier, e.g., a large-sized air purifier that is used in indoor space. Since the portable air purifier is frequently gripped by the user, for use, the gap between the fan **5450** and the bell mouth **5463** often decreases as the portable air purifier is pressed in the front-rear direction.

(301) It is inappropriate to decrease the gap between the bell mouth **5463** and the shroud **5457**, as a way of reducing the effect of whirlwind.

(302) Considering the fact, in the embodiment, the fan base **6460** is provided with a projection rib **6468**, as illustrated in FIGS. 18 and 21.

(303) The projection rib **6468**, as described above, may protrude in a way that forms a slant surface. The slant surface of the projection rib **6468** is spaced a predetermined distance apart from the shroud **5457** and formed in parallel with the slant surface of the shroud **5457**.

(304) The projection rib **6468** may be provided in a way that extends in the centrifugal direction from the boundary portion between the bell mouth **6463** and the base plate **5461**. Accordingly, a gap between the fan base **6460** and the shroud **5457** may remain constant in the portion where the projection rib **6468** is formed. Accordingly, a gap is formed between the projection rib **6468** and the shroud **5457**, in the portion where the projection rib **6468** is formed, and the length of the gap in the axial direction may remain constant within the area where the projection rib **6468** is formed.

(305) That is, in the area where the projection rib **6468** is formed, the width of the gap between the fan base **6460** and the shroud **5457** may remain constant without increasing toward the centrifugal

direction (P2-P3).

(306) Additionally, the entrance of the gap, i.e., the width P3 of the outermost portion of the gap in the radial direction of the gap, which is open in the lateral direction, may be maintained at the width of the gap or less at the minimum gap point P1. That is, unlike the above-described structure, an environment where whirlwind hardly flows into the fan module through the gap between the bell mouth **6463** and the shroud **5457** may be created, in the embodiment.

(307) Accordingly, even if the fan base **6460** and the fan **5450** do not contact each other excessively to reduce the gap between the bell mouth **6463** and the shroud **5457**, the generation of whirlwind may be suppressed effectively.

(308) Thus, the occurrence of flow loss caused by whirlwind may be suppressed, and the fan base **6460** and the fan **5450** do not need to contact each other excessively, reducing the possibility of failure in a product, caused by difficulty in assembly and tolerance.

(309) Further, the projection rib **6468** may also help to increase the thickness of the peripheral portion of the bell mouth **6463**. Accordingly, even if an external force is applied to the portable air purifier, the bell mouth **6463** and its peripheral portion are rarely deformed. That is, although the portable air purifier is pressed in the front-rear direction, the gap between the fan **5450** and the bell mouth **6463** is not narrowed easily.

(310) As a result, the possibility of contact between the fan **5450** and the bell mouth **6463**, caused by an external force, may decrease, and wear or damage to the fan **5450** and its surrounding components, caused by contact between the fan **5450** and the bell mouth **6463**, is much less likely to occur.

(311) Furthermore, as resistance against the deformation of the bell mouth **6463** and its peripheral portion increases, the gap between the fan base **6460** and the shroud **5457** may be designed to decrease further. Accordingly, the possibility of the occurrence of flow loss caused by whirlwind may decrease further.

(312) FIG. 22 is a table showing results of measurement of noise and press volume, with or without a projection rib.

(313) FIG. 22 shows results of experiments of a portable air purifier including a fan base without a projection rib, and the portable air purifier in the embodiment, i.e., a portable air purifier including a fan base with a projection rib.

(314) The experiments were performed under the same conditions except that the projection rib was applied or not. In relation to the portable air purifier without the projection rib, experiments were performed on the condition that the widths of the minimum gap point were 0.4 mm and 1.0 mm. In relation to the portable air purifier with the projection rib, an experiment was performed on the condition that the width of the minimum gap point was 1 mm.

(315) Additionally, noise was measured on condition that air volume was the same. Increasing noise may denote increasing flow loss. Accordingly, results of the measurement of noise may help to indirectly determine whether whirlwind is formed.

(316) Further, to measure the press volume, a force was applied to the front of the portable air purifier with a pushpull gauge, and a numerical value measured by the pushpull gauge was checked when noise was made because of contact between the fan and the fan base.

(317) Hereafter, the effect of the portable air purifier in the embodiment is described with reference to FIGS. 21 and 22.

(318) The results of the experiments indicate that excellent noise performance was ensured (41.7 dB) while the press volume was low (4.7N), at the 0.4-mm width of the minimum gap point P1, without the projection rib (the first experimental example).

(319) The results of the experiments indicate that noise increased (42.2 dB) while the press volume increased (27.9N), at the 1-mm width of the minimum gap point P1, without the projection rib (the second experimental example).

(320) In the above experiments, a decreasing width of the minimum gap point P1 resulted in

decreasing noise, without the projection rib. However, the fan **5450** and the bell mouth **6463** contacted each other easily. An increasing width of the minimum gap point P1 resulted in increasing noise.

(321) In the third experiment with the projection rib **6468**, noise performance was 41.7 dB similar to that of the first experimental example, and the press volume was 28.4N higher than that of the second experimental example.

(322) The results of the experiments are as follows.

(323) That is, the portable air purifier **50** with the projection rib **6468** in the embodiment may effectively suppress the occurrence of flow loss caused by whirlwind, and the contact between the fan **5450** and the fan base **6460** caused by an external force.

(324) The portable air purifier **50** in the embodiment provides a high-level of suction and discharge performance while suppressing flow loss caused by whirlwind and reducing noise, ensuring improvement in air purification performance.

(325) Additionally, the portable air purifier **50** in the embodiment may effectively suppress contact between the fan **5450** and the fan case **6460**, caused by an external force, reducing the possibility of wear or damage to the fan **5450** and its surrounding components and ensuring improvement in durability.

(326) The embodiments are described above with reference to a number of illustrative embodiments thereof. However, the embodiments set forth herein are provided only as examples, and numerous other modifications and embodiments can be devised by one skilled in the art without departing from the technical spirit of the disclosure. The technical protection scope of the disclosure should be defined according to the appended claims.

DESCRIPTION OF REFERENCE NUMERALS

(327) **50**: Portable air purifier **510**: Front surface panel **510a**: Outlet **520**: Case **525**: First connection surface part **530**: Fan cover **531**: Upper cover part **532**: Upper air discharge part **533**: Lower air discharge part **534**: First fastening projection **535**: Lower cover part **540**, **545**: Fan module **540**: First fan module **545**: Second fan module **550**: Filter module **551**: Filter case **552**: Penetration hole **553**: Second fastening projection **559**: Filter **560**: Battery **570**: Rear surface panel **580**: Rear surface cover **5410**: Shaft **5420**: Motor **5421**: Stator **5423**: Rotor **5430**, **5440**: Fan case **5430**: First support part **5430a**: Outlet **5431**: Support plate **5433**: Boss **5435**: Bearing **5440**: Second support part **5441**: Fastening hole **5442**: Projection boss **5443**: Side rib **5445**: Connection part **5450**: Fan **5451**: Hub **5452**: Hub plate part **5452a**: Shaft coupling part **5453**: Skirt part **5454**: Inner projection part **5455**: Fan blade **5455a**: Leading edge **5455b**: Trailing edge **5455c**: Shroud chord **5455d**: Hub chord **5457**: Shroud **5458**: Shroud projection **5460**, **6460**: Fan base **5461**: Base plate **5462**: Fan inlet **5463**, **6463**: Bell mouth **5465**: Fastening boss **5467**: Third fastening projection **6468**: Projection rib

Claims

1. A portable air purifier, comprising: a case having a suction surface on a first side thereof in an axial direction and having a discharge surface on a second side thereof in the axial direction; a filter disposed between the suction surface and the discharge surface; a fan module disposed between the discharge surface and the filter that suctions air, having passed through the filter, and discharges the air to the discharge surface; and a fan base that is disposed between the filter and the fan module, coupled to the fan module, and forms a passage that connects the filter and the fan module, wherein the fan base comprises: a base plate disposed between the filter and the fan module; and a fan inlet that is formed at the base plate, penetrates the base plate, and forms a passage that connects the filter and the fan module, wherein the fan base further comprises a bell mouth that protrudes from the base plate in the axial direction, wherein the fan inlet is formed at an inside of the bell mouth and penetrates in a radial direction of the bell mouth, wherein the fan module comprises a fan

rotatably disposed between the bell mouth and the discharge surface, wherein the bell mouth has a first surface that faces the filter module and is convex toward the filter module and a second surface that faces the fan module and is depressed, while protruding from an inner circumferential surface of the base plate in the axial direction, wherein at least a portion of the fan is inserted into a space surrounded by the second surface, wherein the fan comprises a hub rotatably provided, a shroud spaced a predetermined distance apart from the hub and disposed outside of the hub in a radial direction of the hub, and a fan blade that protrudes from the hub in a centrifugal direction and connects the hub and the shroud, and wherein at least a portion of the shroud is inserted into the space surrounded by the second surface.

2. The portable air purifier of claim 1, wherein the shroud forms a slanted surface that is inclined in the centrifugal direction as the shroud extends away from the filter module in the axial direction, wherein the fan base further comprises a projection rib that protrudes from the base plate toward the shroud in the axial direction while surrounding the bell mouth from the outside in the radial direction, and wherein the projection rib is spaced a predetermined distance apart from the shroud, and forms a slanted surface parallel with the slanted surface of the shroud.

3. The portable air purifier of claim 2, wherein the slanted surface of the projection rib extends in the centrifugal direction from an end portion outside the bell mouth in the radial direction of the bell mouth.

4. The portable air purifier of claim 3, wherein the base plate, the bell mouth, and the projection rib are integrally formed.

5. The portable air purifier of claim 1, wherein the portable air purifier further comprises a fan cover disposed between the discharge surface and the fan module, wherein the fan cover is provided with an air discharge portion that forms a passage that connects the discharge surface and the fan, wherein the fan module is disposed to face the air discharge portion, wherein the air discharge portion is formed at the fan cover and penetrates in the axial direction, and wherein at least a portion of the fan module is inserted into the air discharge portion.

6. The portable air purifier of claim 5, wherein the fan base further comprises a bell mouth that protrudes from the base plate in the axial direction, and wherein at least a portion of the bell mouth is disposed to overlap the fan module.
